Automated Pentest Report Generation Tool

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1.0 Executive Summary

The AppSec team currently faces significant overhead in manually creating penetration test reports, which involves compiling information about vulnerabilities, their severity, impact, and recommended mitigations, along with associated evidence.
This project proposes the development of an automated pentest report generation tool that will streamline this process, reduce manual effort, and increase efficiency by automating the population of commonly found details in the report like vuln description, impact and recommendation, with this tool, the AppSec the tester only need to upload the evidence details and team can focus more on critical analysis and remediation of vulnerabilities and ultimately reducing the reporting time.

1.1 Project Overview

The current process of creating pentest reports is time-consuming and labor-intensive for the AppSec team. It involves a redundant process of manually compiling vulnerability descriptions, severities, impacts, and recommendations, along with associated evidence. This project aims to automate the report creation process, thereby reducing overhead and increasing efficiency**.**

1.2 Scope

1. Focus on automating report creation for web application pentests.
2. Support a vulnerability database of various vulnerability types such as SQL injection, XSS, and Authentication Bypass.
3. Generate standardized reports for internal and client use.

1.3. Objectives

4. Develop a web-based application to automate pentest report generation.
5. Implement a database to store detailed information about vulnerabilities so that they can be re-used for other applications on the go.
6. Create a user-friendly interface for uploading vulnerability evidence.
7. Automate the report generation process using pre-defined templates.
8. Ensure security and data privacy throughout the process.

1.4 Tools and Technology

The technology stack consists of the use of the following -

ID	Architecture	Technology
1	**WebApp**	Django for building the user interface and writing the application logic.
3	**Database**	PostgreSQL for storing vulnerability information and evidence.
4	**File Storage**	Local server storage or similar for storing uploaded evidence.
5	**Document Generation**	Python-Docx library for creating editable Word documents.

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1.5. Timeline

Week 1: Project Setup and Basic Functionality

Week 1, Days 1-2

1. **Project Initialization**
   - Set up the Django project and application structure.
   - Initialize the project repository on GitHub or GitLab.
   - Install required libraries (django, python-docx).

2. **Working with Template**
   - Interacting with the Word document template via docx library.
   - Identify placeholders for dynamic content (e.g., <Executive_Summary>, <Project_Overview>).

3. **Basic Document Manipulation**
   - Write initial scripts to load the template and replace placeholders with static text.

Test basic text replacement and table updates using python-docx

Week 1, Days 3-5

4. **Data Structure Design**
   - Design the data models to store information about vulnerabilities and other report sections.

Week 2: Database Integration and Advanced Functionality

Week 2, Days 1-3

1. **Database Setup**
   - Configure the database settings in settings.py.
   - Create and run migrations for the vulnerability model.

2. **Database Integration**
   - Write views and forms to add, edit, and delete vulnerabilities.
   - Integrate database queries with the document generation script.

3. **Dynamic Content Population**
   - Enhance the script to dynamically populate the report with data fetched from the database.

Week 2, Days 4-5

4. **Error Handling and Logging**
   - Implement comprehensive error handling for major operations.
   - Add logging to capture detailed error information for troubleshooting.

Week 3: UI Development and Integration

Week 3, Days 1-3

1. **UI Design**
   - Design the UI using Django templates.
   - Create forms and views for user input.

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2. **UI Implementation**
    - Develop the basic UI layout and components.
    - Implement form fields to capture user inputs (e.g., project overview, assessment team details).

Week 3, Days 4-5

3. **UI and Backend Integration**
    - Connect the UI with the backend script to trigger report generation based on user inputs.
    - Ensure that user inputs are validated and sanitized.

Week 4: Final Testing, Deployment, and Documentation

Week 4, Days 1-3

1. **Comprehensive Testing**
    - Perform thorough testing with various scenarios to ensure all functionalities work as expected.
    - Conduct performance testing to ensure the system handles large datasets efficiently.
2. **User Acceptance Testing (UAT)**
    - Share the application with a small group of users for feedback.
    - Collect and implement feedback to improve the application.

Week 4, Days 4-5

3. **Documentation**
    - Prepare comprehensive documentation covering the setup, usage, and troubleshooting of the application.
    - Include code comments and inline documentation for maintainability.
4. **Deployment**
    - Deploy the application to a production environment.
    - Ensure all dependencies are correctly installed and configured.

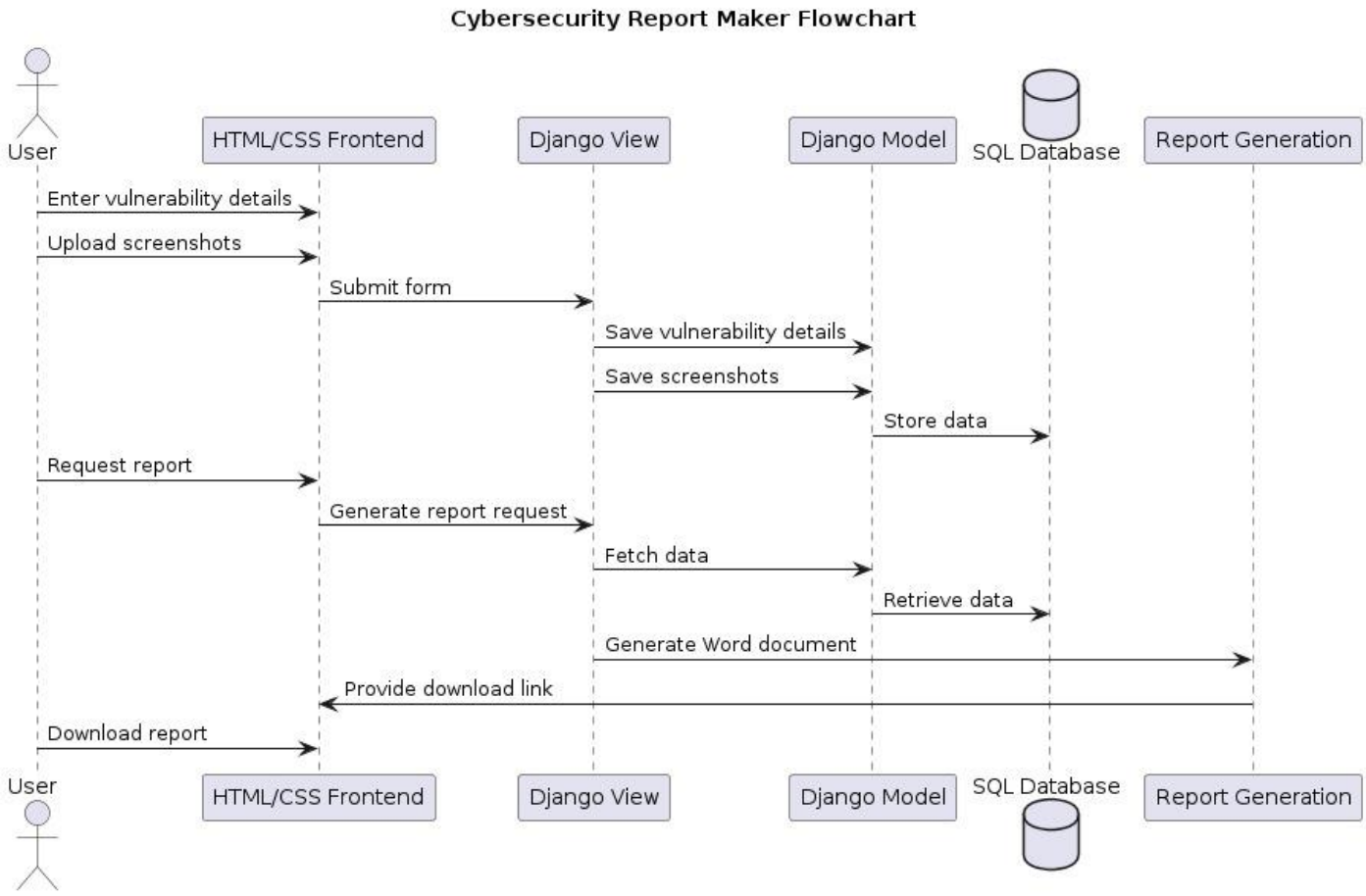
5. **Final Review and Launch**
    - Conduct a final review with stakeholders.
    - Officially launch the application.

This timeline provides a structured approach to developing the tool and provides a high level view of deliverables.

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1.6 Project Workflow



1.7 Security and Privacy Considerations

- Ensure all data is encrypted both in transit and at rest.
- Implement secure authentication and access control mechanisms.
- Regularly update and patch all software dependencies.

1.8 Conclusion

Automating the pentest report creation process will allow the AppSec team to focus more on analyzing and addressing vulnerabilities, leading to improved efficiency and more robust security assessments.

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